



UNIVERSITY MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE & MATHEMATICS

SOFTWARE ARCHITECTURE
CSE3433

INDIVIDUAL LOGBOOK

GROUP NO. 11

SERVICE-ORIENTED SMART CAMPUS SUPPORT SYSTEM

PREPARED BY:
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PREPARED FOR:
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[BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING)
WITH HONOURS]
SEMESTER 2 2025/2026

1.0 Team Information

Name	Matric No	Role	Task
Elsa Fadlin Binti Mersing	S75412	Frontend Developer & Tester	• Design and develop the user interface • Ensure system is user-friendly and responsive • Conduct system testing and debugging • Report and fix errors to improve system quality
Sarah Amira Heri	S75812	Project Manager & System Analyst	• Plan and manage the overall project timeline • Gather and analyze system requirements • Define system architecture (SOA design) • Ensure the project meets objectives
Siti Aisyah Arwiah Binti Yundi	S75256	Backend Developer & Database Designer	• Develop core system functionalities (services in SOA)

			• Handle server-side logic and integration between services • Design and manage the database structure • Ensure data consistency and security
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2.0 Weekly Progress Report

Week 4

Item	Description
Week	Week 4 (2 - 8 March 2026)
Weekly Activities	Brainstormed the foundational service-oriented architecture (SOA) components for the Smart Campus Support System.
Individual Contributions	Conducted requirements gathering for Notification Module. Made the first draft for course creation.
Challenges	Determine the loose coupling between the services.
Action Plans	Conducted research regarding loose coupling between the services in SOA.

Week 5

Item	Description
Week	Week 5 (9 - 15 March 2026)
Weekly Activities	Drafted the first deliverable project proposal report for submission on 16 th April 2026.
Individual Contributions	Tasked with problem description and explain the selected architecture style of the project. Wrote the full format for the project.
Challenges	Managing structural interface designs before backend service endpoints are officially established.
Action Plans	Review with lecturer on the proposal.

Week 6

Item	Description
Week	Week 6 (16 - 22 March 2026)
Weekly Activities	Finalized the project proposal report.
Individual Contributions	Contributed to System Decomposition by structuring a Component Table defining frontend single-responsibility UI design patterns.

Challenges	Ensuring clear component boundaries to split dense presentation views from foundational backend business micro-services cleanly.
Action Plans	Collaborate with the group's Systems Analyst to explicitly frame Boundary Statements for each autonomous sub-service block.

Week 7

Item	Description
Week	Week 7 (23 - 29 March 2026)
Weekly Activities	Designed the preliminary SOA architecture.
Individual Contributions	Developed a high-level architecture diagram showing service interactions.
Challenges	Defining communication mechanisms between services.
Action Plans	Research service integration techniques and messaging approaches.

Week 8

Item	Description
Week	Week 8 (30 March - 5 April 2026)
Weekly Activities	Evaluated architectural alternatives and finalized SOA adoption.
Individual Contributions	Reviewed SOA design principles and refined the service interaction diagram.
Challenges	Selecting the most suitable architecture based on project requirements.
Action Plans	Document architectural justifications and design decisions.

Week 9

Item	Description
Week	Week 9 (6 - 12 April 2026)
Weekly Activities	Developed UML diagrams for the architecture design.

Individual Contributions	Created System Context Diagram, Component Diagram, and Interaction Diagram.
Challenges	Maintaining consistency between diagrams and requirements.
Action Plans	Validate UML diagrams with team members and supervisor.

Week 10

Item	Description
Week	Week 10 (13 - 19 April 2026)
Weekly Activities	Designed database structure and service integration workflow.
Individual Contributions	Defined entities, relationships, and service data flow.
Challenges	Ensuring services can exchange data efficiently while remaining loosely coupled.
Action Plans	Prepare database schema and integration strategy documentation.

Week 11

Item	Description
Week	Week 11 (20 - 26 April 2026)
Weekly Activities	Prepared prototype implementation plan.
Individual Contributions	Documented module structure, service responsibilities, and implementation scope.
Challenges	Translating architecture design into a practical implementation plan.
Action Plans	Begin developing core system modules according to SOA design.

Week 12

Item	Description
Week	Week 12 (27 April – 3 May 2026)
Weekly Activities	Implemented system prototype and service components.

Individual Contributions	Assisted in developing JSP pages and database connectivity modules.
Challenges	Integrating frontend interfaces with backend services and database operations.
Action Plans	Conduct preliminary testing and fix identified implementation issues.

Week 13

Item	Description
Week	Week 13 (4 – 10 May 2026)
Weekly Activities	Finished Project Observation 2 and compiled all layout assets for the group submission report.
Individual Contributions	Generated system interaction screenshots, interface navigation flow charts, and repository file maps for inclusion in the report.
Challenges	Formatting all team member content sections into a single document layout cleanly.
Action Plans	Complete final checks and assemble the individual logbook.

3.0 GenAI Usage Record

Week	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	ChatGPT	Understand software architecture concepts	"What are the fundamental principles of software architecture?"	Identified architecture principles, quality attributes, and architectural styles.	Asked for explanations of core architecture concepts and principles.
Week 3	ChatGPT	Explore Architecture Patterns	"What architecture patterns are commonly used in software development?"	Identified Layered, SOA, Microservices, and Event-Driven Architecture patterns.	Requested comparisons between common architecture patterns.
Week 4	Gemini	Requirement Analysis	"Can you help me explain where the non-functional requirements in the system?"	Produced a structured explanation requirements list.	Used AI to determine system requirements.
Week 4	Gemini	Service Identification	"How can I divide my system into separate services using SOA?"	Identified candidate services and responsibilities.	Determined possible services in the system.

Week 5	ChatGP T	Learn SOA Principles	"Can you explain the main principles of Service- Oriented Architecture (SOA)?"	Learned loose coupling, reusability, autonomy, and interoperability.	Requested explanations of SOA concepts.
Week 5	Gemini	Service Boundary Design	"How do I determine the boundaries between services in an SOA system?"	Established clearer service responsibilities.	Defined service responsibilities.
Week 6	Gemini	Architectural Design	"Can you suggest an SOA architecture design for my system?"	Produced a high- level architecture structure.	Defined service responsibilities .
Week 6	Gemini	Quality Attribute Analysis	"How do scalability, security, and reliability affect SOA design?"	Identified important quality attributes.	Examined system quality requirements.
Week 7	ChatGP T	Architecture Comparison	"What is the difference between SOA and	Compared architectural approaches and trade-offs.	Evaluated architecture alternatives.

			Microservices Architecture?"		
Week 7	Gemini	Service Communication	"How do services communicate with each other in SOA?"	Understood synchronous and asynchronous communication.	Learned how services communicate.
Week 7	Gemini	Integration Technology Research	"Can you explain how Apache Kafka works in distributed systems?"	Learned message brokering and event communication concepts.	Investigated messaging technologies.
Week 8	Gemini	Architecture Selection	"Which architecture pattern is most suitable for my project and why?"	Chose an architecture based on project requirements.	Selected the most suitable architecture.
Week 8	Gemini	UML Diagram Support	"What UML diagrams should I include in my software architecture project?"	Identified Use Case, Class, and Sequence Diagrams.	Created architecture-related diagrams.
Week 9	Gemini	Dynamic Modelling	"Can you help me create a sequence	Produced interaction flow between components.	Developed system

			diagram for my system?"		interaction flows.
Week 9	Gemini	Business Perspective Analysis	"Can you explain the revenue streams for my software system?"	Generated descriptions for revenue models.	Explored revenue opportunities.
Week 10	ChatGP T	Partnership Analysis	"Who should be the key partners for my software system?"	Identified cloud, payment, and API providers.	Identified external stakeholders.
Week 10	Gemini	API Planning	"What APIs should I integrate into my system?"	Recommended payment, location, and notification APIs.	Planned external integrations.
Week 10	ChatGP T	Data Architecture Support	"How should my database entities be distributed across services in SOA?"	Suggested data ownership and service-specific data boundaries.	Used AI to review database and service relationships.
Week 11	Gemini	Documentation Support	"Can you help me explain the fundamental principles of software	Produced report-ready explanations.	Improved report content.

			architecture for my report?"		
Week 11	Gemini	Prototype Planning	"What should I write in the prototype implementation section?"	Generated prototype documentation content.	Defined implementation scope.
Week 12	Gemini	Development Guidance	"How should I structure my JSP application for this project?"	Produced implementation guidance and architecture alignment.	Assisted during implementation.
Week 12	Gemini	Database Integration	"How do I connect my database in JSP using JDBC?"	Established database connectivity approach.	Connected application components.
Week 12	ChatGPT	Prototype Evaluation	"Does my prototype implementation reflect the architecture design requirements?"	Produced recommendations for alignment between design and implementation.	Used AI to assess implementation progress.
Week 13	Gemini	Testing Support	"Can you help me create test cases for my system?"	Produced functional testing scenarios.	Prepared validation activities.

Week 13	Gemini	Final Documentation	"Can you review and improve my software architecture report?"	Improved report quality and presentation readiness.	Refined final deliverables.
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